Model Comparison: Accuracy + Confusion Matrices

Models compared:

1. VADER (lexicon)
2. TF-IDF + Logistic Regression
3. Hybrid SVM + Lexicon
4. FinBERT Baseline
5. FinBERT Finetuned ← best model

Install & Imports

```
!pip install transformers torch scikit-learn vaderSentiment seaborn
matplotlib joblib --quiet
import nltk
nltk.download('vader_lexicon', quiet=True)
```

```
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━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 126.0/126.0 kB 4.7 MB/s eta
0:00:00
```

```
True
```

```
import pandas as pd
import numpy as np
import torch
import joblib
import re
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, classification_report,
    confusion_matrix, ConfusionMatrixDisplay
)
from nltk.sentiment import SentimentIntensityAnalyzer
from transformers import AutoTokenizer,
AutoModelForSequenceClassification, pipeline
```

```
DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
LABELS = ['negative', 'neutral', 'positive']
print(f'Device: {DEVICE}')
```

```
Device: cuda
```

1. Load & Split Data

Uses the **same** balanced 9000-row dataset and same random seed as your original notebook so results are directly comparable.

```
df = pd.read_csv('articles.csv')
df.columns = [c.strip() for c in df.columns]

# Balance to 3000 per class (same as original)
df_balanced = (
    df.groupby('Sentiment', group_keys=False)
      .apply(lambda x: x.sample(min(len(x), 3000), random_state=42))
      .reset_index(drop=True)
)
df_balanced['Sentiment'] =
df_balanced['Sentiment'].str.strip().str.lower()

# Same 70/15/15 split as original
X_train, X_temp, y_train, y_temp = train_test_split(
    df_balanced['Sentence'], df_balanced['Sentiment'],
    test_size=0.30, random_state=42, stratify=df_balanced['Sentiment']
)
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp,
    test_size=0.50, random_state=42, stratify=y_temp
)

print(f'Train: {len(X_train)} | Val: {len(X_val)} | Test:
{len(X_test)}')
print(pd.Series(y_test).value_counts())
```

```
Train: 3998 | Val: 857 | Test: 857
Sentiment
neutral     450
positive    278
negative    129
Name: count, dtype: int64
```

Helper: Unified Evaluation Function

```
# Store results for final comparison
all_results = {}

def evaluate_model(name, y_true, y_pred, color='Blues'):
```

```
    acc = accuracy_score(y_true, y_pred)
    report = classification_report(y_true, y_pred, labels=LABELS,
output_dict=True)
    cm = confusion_matrix(y_true, y_pred, labels=LABELS)

    # Store for final comparison chart
    all_results[name] = {
        'accuracy': acc,
        'f1_macro':    report['macro avg']['f1-score'],
        'f1_negative': report['negative']['f1-score'],
        'f1_neutral':  report['neutral']['f1-score'],
        'f1_positive': report['positive']['f1-score'],
    }

    # Print text report
    print('=' * 55)
    print(f'  {name}')
    print('=' * 55)
    print(f'  Accuracy : {acc:.4f} ({acc*100:.2f}%)')
    print(f'  Macro F1 : {report["macro avg"]["f1-score"]:.4f}')
    print()
    print(classification_report(y_true, y_pred, labels=LABELS))

    # Confusion matrix — accuracy in title, NOT overlaid on the chart
    fig, ax = plt.subplots(figsize=(6, 5))
    sns.heatmap(
        cm, annot=True, fmt='d', cmap=color,
        xticklabels=LABELS, yticklabels=LABELS,
        linewidths=0.5, linecolor='grey', ax=ax
    )
    ax.set_title(
        f'Confusion Matrix — {name}\nAccuracy: {acc*100:.2f}%',
        fontsize=12, fontweight='bold', pad=14
    )
    ax.set_xlabel('Predicted Label', fontsize=11)
    ax.set_ylabel('True Label', fontsize=11)
    plt.tight_layout()
    plt.show()
    return acc

print('evaluate_model() ready')
```

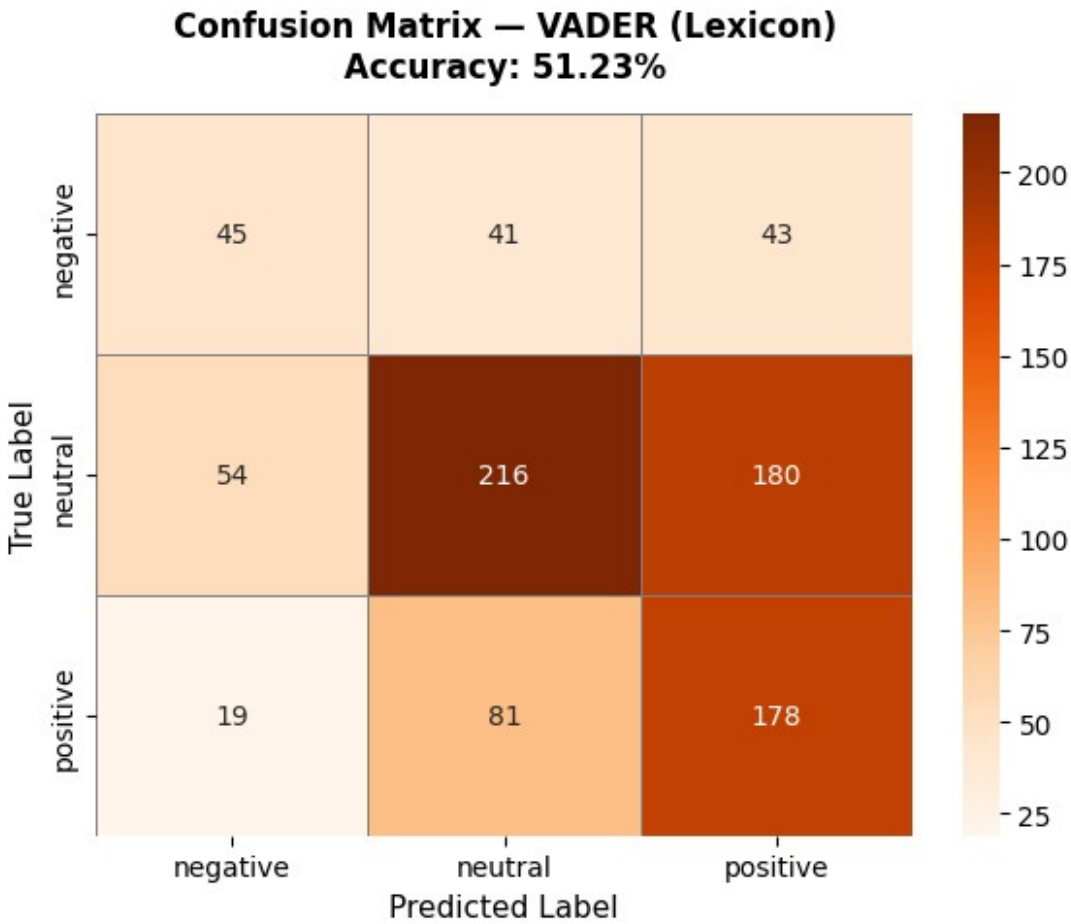
```
evaluate_model() ready
```

VADER (Lexicon)

```
sia = SentimentIntensityAnalyzer()

def vader_label(text, pos_thresh=0.10, neg_thresh=-0.10):
    compound = sia.polarity_scores(text)['compound']
```

```
    if compound >= pos_thresh:   return 'positive'
    elif compound <= neg_thresh: return 'negative'
    else:                        return 'neutral'

y_pred_vader = [vader_label(t) for t in X_test]
evaluate_model('VADER (Lexicon)', y_test, y_pred_vader,
color='Oranges')
```

```
==========================================================
  VADER (Lexicon)
==========================================================
  Accuracy : 0.5123 (51.23%)
  Macro F1 : 0.4790

              precision    recall  f1-score   support

    negative       0.38      0.35      0.36       129
     neutral       0.64      0.48      0.55       450
    positive       0.44      0.64      0.52       278

    accuracy                           0.51       857
   macro avg       0.49      0.49      0.48       857
weighted avg       0.54      0.51      0.51       857
```

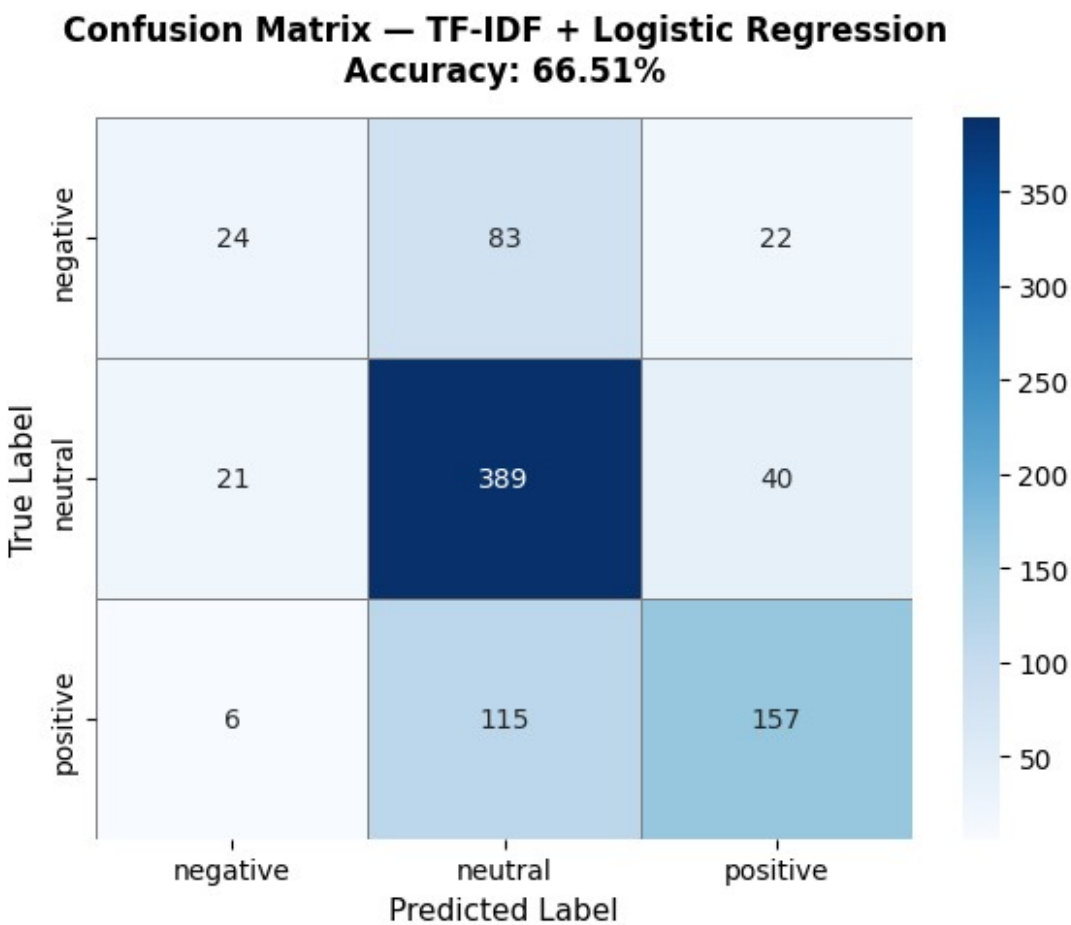



```
0.5122520420070011
```

TF-IDF + Logistic Regression

```
lr_pipe = Pipeline([
    ('tfidf', TfidfVectorizer(
        lowercase=True, stop_words='english',
        ngram_range=(1, 2), min_df=5, max_df=0.9
    )),
    ('logreg', LogisticRegression(
        max_iter=2000, multi_class='multinomial',
        solver='lbfgs', n_jobs=-1
    ))
])
lr_pipe.fit(X_train, y_train)

y_pred_lr = lr_pipe.predict(X_test)
evaluate_model('TF-IDF + Logistic Regression', y_test, y_pred_lr,
color='Blues')
```

```
============================================================
  TF-IDF + Logistic Regression
============================================================
  Accuracy : 0.6651 (66.51%)
  Macro F1 : 0.5496

              precision    recall  f1-score   support

    negative       0.47      0.19      0.27       129
     neutral       0.66      0.86      0.75       450
    positive       0.72      0.56      0.63       278

    accuracy                           0.67       857
   macro avg       0.62      0.54      0.55       857
weighted avg       0.65      0.67      0.64       857
```



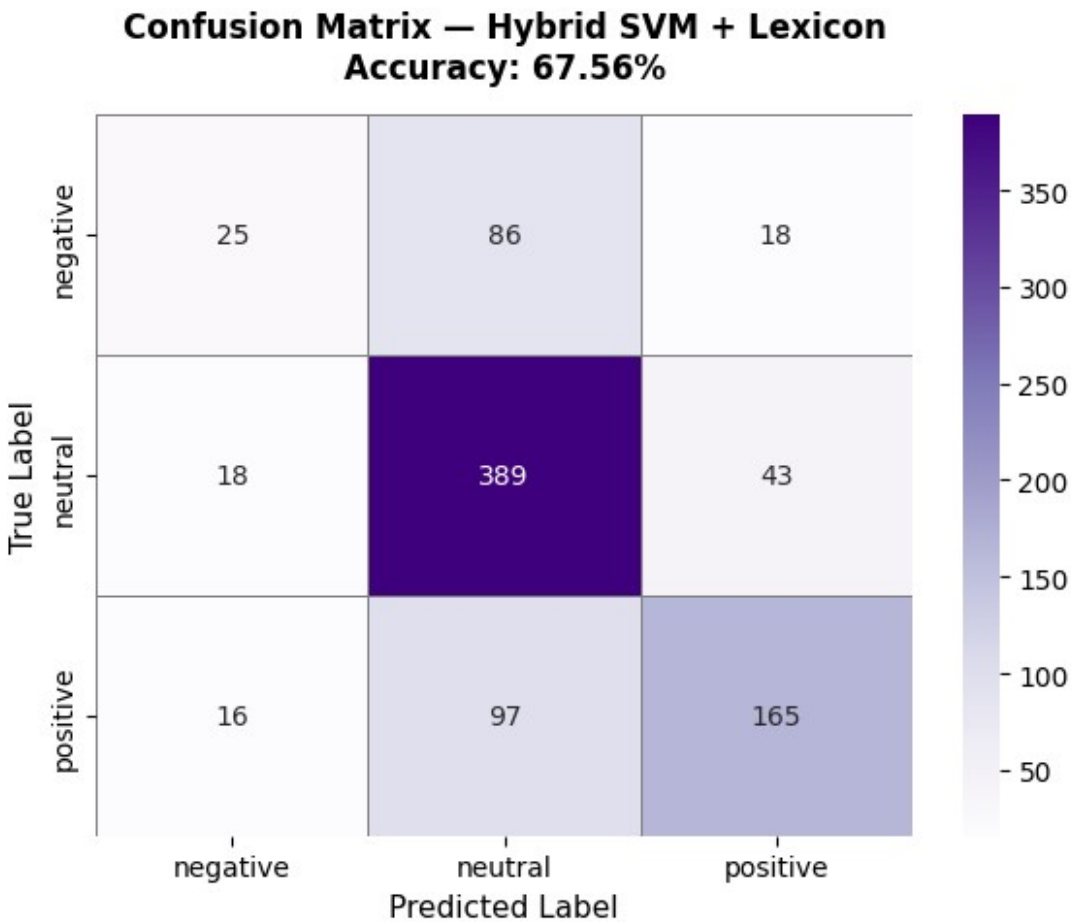

```
0.6651108518086347
```

Hybrid SVM + Lexicon

```
svm_pipe = Pipeline([
    ('tfidf', TfidfVectorizer(
        stop_words='english', ngram_range=(1, 2),
        min_df=3, max_df=0.9
    )),
    ('svm', SVC(kernel='linear', probability=True))
])
svm_pipe.fit(X_train, y_train)

# Keyword boost (same as your original)
strong_neg_words = {
    'crashed', 'crash', 'plummeted', 'plummet', 'losses', 'loss',
    'fell sharply', 'tumbled', 'panic', 'panicking', 'collapsed',
    'scandal', 'bankrupt', 'bankruptcy'
}
strong_pos_words = {
    'smashed expectations', 'beat expectations', 'soared', 'jumped',
    'surged', 'record profits', 'strong growth', 'optimistic',
    'bullish', 'rallied', 'rebounded'
}

def svm_with_lexicon_boost(text, boost=0.5):
    base_proba = svm_pipe.predict_proba([text])[0]
    classes = svm_pipe.classes_
    prob_dict = dict(zip(classes, base_proba))
    t = text.lower()
    if any(w in t for w in strong_neg_words):
        prob_dict['negative'] += boost
    if any(w in t for w in strong_pos_words):
        prob_dict['positive'] += boost
    total = sum(prob_dict.values())
    prob_dict = {k: v/total for k, v in prob_dict.items()}
    return max(prob_dict, key=prob_dict.get)

print('Running Hybrid SVM predictions (may take a minute)...')
y_pred_svm = [svm_with_lexicon_boost(t) for t in X_test]
evaluate_model('Hybrid SVM + Lexicon', y_test, y_pred_svm,
color='Purples')
```

```
Running Hybrid SVM predictions (may take a minute)...
============================================================
  Hybrid SVM + Lexicon
============================================================
  Accuracy : 0.6756 (67.56%)
  Macro F1 : 0.5607

              precision    recall  f1-score   support
```

```
    negative       0.42      0.19      0.27       129
     neutral       0.68      0.86      0.76       450
    positive       0.73      0.59      0.65       278

    accuracy                           0.68       857
   macro avg       0.61      0.55      0.56       857
weighted avg       0.66      0.68      0.65       857
```




```
0.67561260210035
```

FinBERT Baseline

```
print('Loading FinBERT Baseline...')
fb_tokenizer = AutoTokenizer.from_pretrained('ProsusAI/finbert')
fb_model =
AutoModelForSequenceClassification.from_pretrained('ProsusAI/finbert')

label_map = {0: 'positive', 1: 'negative', 2: 'neutral'}
fb_model.config.id2label = label_map
```

```
fb_model.config.label2id = {v: k for k, v in label_map.items()}

finbert_pipe = pipeline(
    'text-classification', model=fb_model, tokenizer=fb_tokenizer,
    device=0 if torch.cuda.is_available() else -1,
    truncation=True, max_length=512, batch_size=32
)

print(f'Running FinBERT Baseline on {len(X_test)} samples...')
preds = finbert_pipe(X_test.tolist())
y_pred_fb_base = [p['label'].lower().strip() for p in preds]
evaluate_model('FinBERT Baseline', y_test, y_pred_fb_base,
color='YlOrRd')
```

```
Loading FinBERT Baseline...
```

{"model_id":"5539ea4756374cd3ad79954ee9b64875","version_major":2,"version_minor":0}

```
Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.
```

{"model_id":"83b7a8d97f564fcba13cfcbfa3e15939","version_major":2,"version_minor":0}

{"model_id":"88e630c57df94923bfa7cd87a372bdf4","version_major":2,"version_minor":0}

{"model_id":"34e0fa0ed35e4fb2b0f4f78dec346750","version_major":2,"version_minor":0}

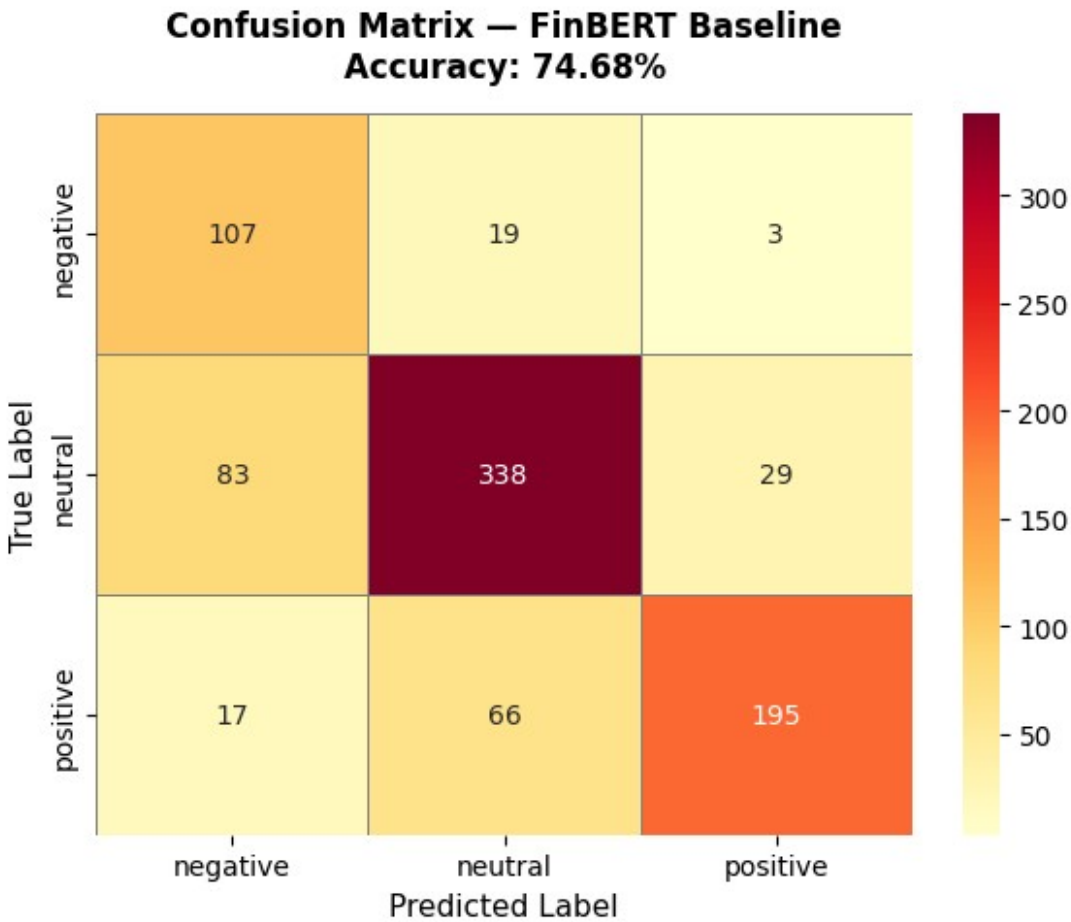
{"model_id":"f3d3c7aad41b48dfbfe031ad45ca106d","version_major":2,"version_minor":0}

{"model_id":"4f0ea7b84b0a4c56af29e8692b45ff1a","version_major":2,"version_minor":0}

{"model_id":"7385926cc3064ea1ac0b016b87a3f122","version_major":2,"version_minor":0}

```
BertForSequenceClassification LOAD REPORT from: ProsusAI/finbert
Key                           | Status     |  |
------------------------------+------------+--+-
bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.
```

```
Running FinBERT Baseline on 857 samples...
============================================================
  FinBERT Baseline
============================================================
  Accuracy : 0.7468 (74.68%)
  Macro F1 : 0.7278

              precision    recall  f1-score   support

    negative       0.52      0.83      0.64       129
     neutral       0.80      0.75      0.77       450
    positive       0.86      0.70      0.77       278

    accuracy                           0.75       857
   macro avg       0.72      0.76      0.73       857
weighted avg       0.78      0.75      0.75       857
```




```
0.7467911318553092
```

FinBERT Finetuned

```
print('Loading FinBERT Finetuned...')
ft_tokenizer = AutoTokenizer.from_pretrained('project-aps/finbert-finetune')
ft_model = AutoModelForSequenceClassification.from_pretrained('project-aps/finbert-finetune')

label_map_ft = {0: 'neutral', 1: 'negative', 2: 'positive'}
ft_model.config.id2label = label_map_ft
ft_model.config.label2id = {v: k for k, v in label_map_ft.items()}

finbert_ft_pipe = pipeline(
    'text-classification', model=ft_model, tokenizer=ft_tokenizer,
    device=0 if torch.cuda.is_available() else -1,
    truncation=True, max_length=512, batch_size=32
)

print(f'Running FinBERT Finetuned on {len(X_test)} samples...')
preds_ft = finbert_ft_pipe(X_test.tolist())
y_pred_fb_ft = [p['label'].lower().strip() for p in preds_ft]
evaluate_model('FinBERT Finetuned', y_test, y_pred_fb_ft, color='Greens')

# ── Save FinBERT Finetuned model ──────────────────────────────────────────
import os
os.makedirs('finbert_finetuned', exist_ok=True)
ft_model.save_pretrained('finbert_finetuned')
ft_tokenizer.save_pretrained('finbert_finetuned')
print('□ FinBERT Finetuned saved to finbert_finetuned/')

# Zip and download
!zip -r finbert_finetuned.zip finbert_finetuned/
from google.colab import files
files.download('finbert_finetuned.zip')
print('□ finbert_finetuned.zip downloaded!')
```

```
Loading FinBERT Finetuned...
```


```
{"model_id":"9fe18eccdbc8439f867bbe326709b3a5","version_major":2,"version_minor":0}

{"model_id":"f44f6018e14c4b03a33d8bddee1a0207","version_major":2,"version_minor":0}

{"model_id":"375dc8d17946485aa7632dab297e88e5","version_major":2,"version_minor":0}
```

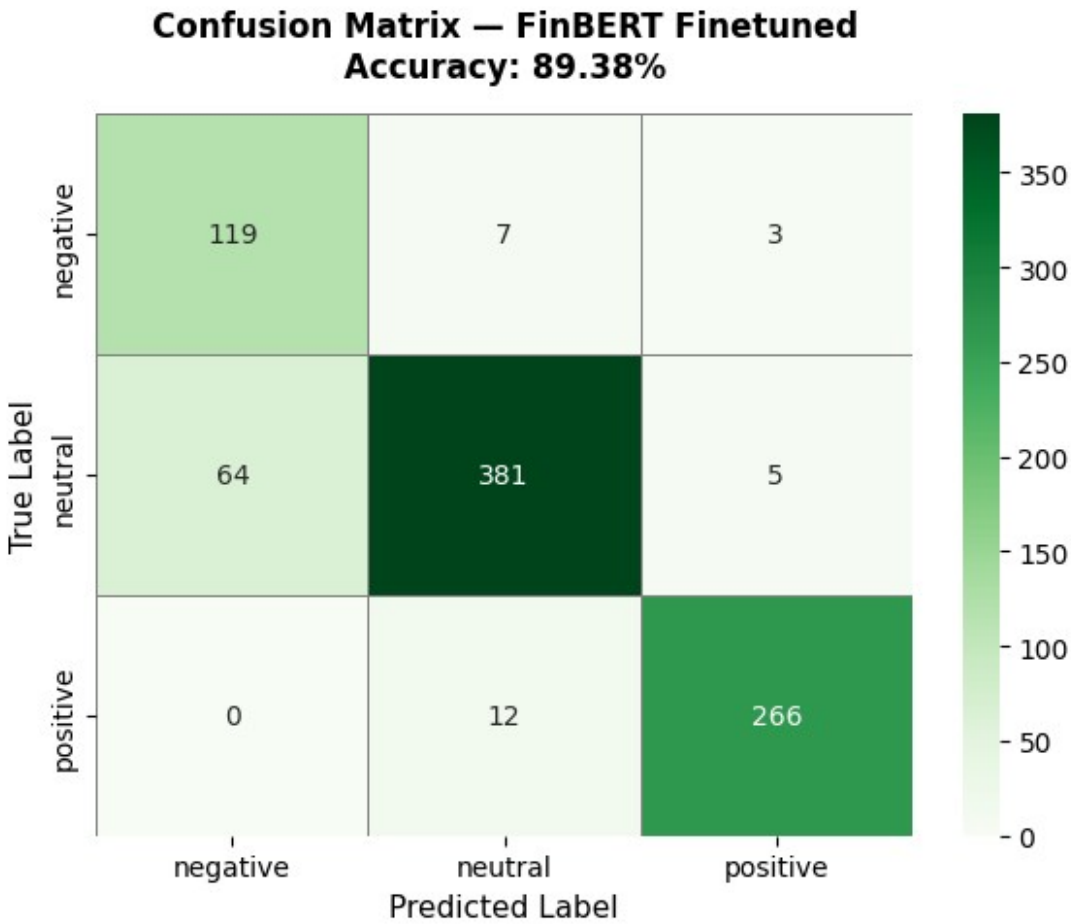
{"model_id":"64074e70ec484ac68af65898b79d2d54","version_major":2,"version_minor":0}

{"model_id":"b5de8db30f4549338e53745297127817","version_major":2,"version_minor":0}

{"model_id":"3bce7c4d468249fb83272c96629968d7","version_major":2,"version_minor":0}

{"model_id":"aa6d03ce372b4e738ca43e3dbbef8f36","version_major":2,"version_minor":0}

```
Running FinBERT Finetuned on 857 samples...
============================================================
  FinBERT Finetuned
============================================================
  Accuracy : 0.8938 (89.38%)
  Macro F1 : 0.8744

              precision    recall  f1-score   support

    negative       0.65      0.92      0.76       129
     neutral       0.95      0.85      0.90       450
    positive       0.97      0.96      0.96       278

    accuracy                           0.89       857
   macro avg       0.86      0.91      0.87       857
weighted avg       0.91      0.89      0.90       857
```




```
{"model_id":"92292e36ecfc411abd73f0d55476d560","version_major":2,"version_minor":0}

 FinBERT Finetuned saved to finbert_finetuned/
  adding: finbert_finetuned/ (stored 0%)
  adding: finbert_finetuned/model.safetensors (deflated 7%)
  adding: finbert_finetuned/tokenizer_config.json (deflated 52%)
  adding: finbert_finetuned/tokenizer.json (deflated 71%)
  adding: finbert_finetuned/config.json (deflated 54%)

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

 finbert_finetuned.zip downloaded!
```

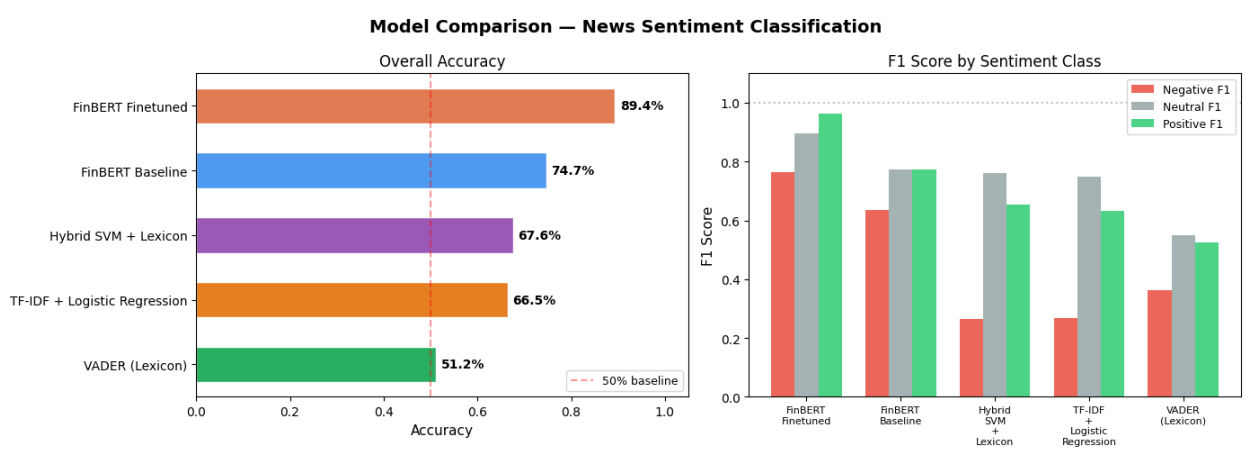
Final Comparison

```
# ── Summary Table ──────────────────────────────────────────────

summary = pd.DataFrame(all_results).T
```

```
summary = summary.sort_values('accuracy', ascending=False)
summary_pct = summary.copy()
for col in summary_pct.columns:
    summary_pct[col] = (summary_pct[col] * 100).round(2).astype(str) +
'%'

print('=' * 65)
print('  MODEL COMPARISON SUMMARY')
print('=' * 65)
print(summary_pct.to_string())
print()

# ── Bar Chart — Accuracy
──────────────────────────────────────────────────────────
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle('Model Comparison — News Sentiment Classification',
fontsize=14, fontweight='bold')

colors = ['#e07b54', '#4e9af1', '#9b59b6', '#e67e22', '#27ae60']
model_names = list(summary.index)
accuracies  = [summary.loc[m, 'accuracy'] for m in model_names]
f1_macros   = [summary.loc[m, 'f1_macro'] for m in model_names]

# Accuracy bar
bars = axes[0].barh(model_names, accuracies, color=colors,
edgecolor='white', height=0.55)
axes[0].set_xlim(0, 1.05)
axes[0].set_xlabel('Accuracy', fontsize=11)
axes[0].set_title('Overall Accuracy', fontsize=12)
for bar, val in zip(bars, accuracies):
    axes[0].text(val + 0.01, bar.get_y() + bar.get_height()/2,
                 f'{val*100:.1f}%', va='center', fontsize=10,
fontweight='bold')
axes[0].axvline(x=0.5, color='red', linestyle='--', alpha=0.4,
label='50% baseline')
axes[0].legend(fontsize=9)
axes[0].invert_yaxis()

# F1 per class grouped bar
x = np.arange(len(model_names))
width = 0.25
f1_neg = [summary.loc[m, 'f1_negative'] for m in model_names]
f1_neu = [summary.loc[m, 'f1_neutral']  for m in model_names]
f1_pos = [summary.loc[m, 'f1_positive'] for m in model_names]

axes[1].bar(x - width, f1_neg, width, label='Negative F1',
color='#e74c3c', alpha=0.85)
axes[1].bar(x,          f1_neu, width, label='Neutral F1',
color='#95a5a6', alpha=0.85)
axes[1].bar(x + width,  f1_pos, width, label='Positive F1',
```

```
color='#2ecc71', alpha=0.85)
axes[1].set_xticks(x)
axes[1].set_xticklabels([m.replace(' ', '\n') for m in model_names],
fontsize=8)
axes[1].set_ylim(0, 1.1)
axes[1].set_ylabel('F1 Score', fontsize=11)
axes[1].set_title('F1 Score by Sentiment Class', fontsize=12)
axes[1].legend(fontsize=9)
axes[1].axhline(y=1.0, color='grey', linestyle=':', alpha=0.5)

plt.tight_layout()
plt.show()
print('Chart saved as model_comparison.png')
```

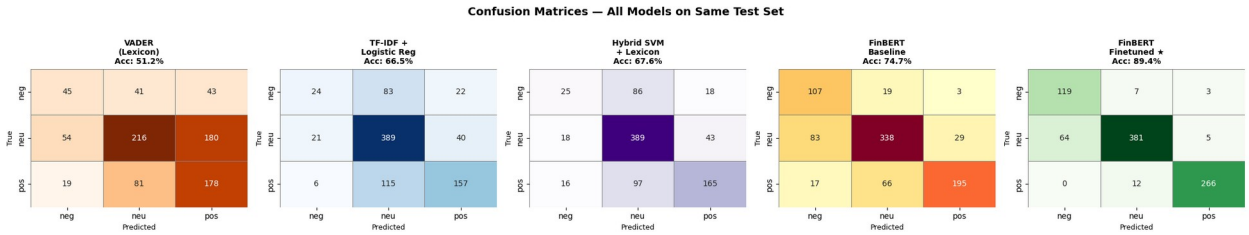
```
====================================================================
  MODEL COMPARISON SUMMARY
====================================================================
                              accuracy f1_macro f1_negative f1_neutral
f1_positive
FinBERT Finetuned               89.38%   87.44%      76.28%     89.65%
96.38%
FinBERT Baseline                74.68%   72.78%      63.69%     77.43%
77.23%
Hybrid SVM + Lexicon            67.56%   56.07%       26.6%     76.13%
65.48%
TF-IDF + Logistic Regression    66.51%   54.96%      26.67%     75.02%
63.18%
VADER (Lexicon)                 51.23%    47.9%      36.44%     54.82%
52.43%
```




```
Chart saved as model_comparison.png
```

Side-by-Side Confusion Matrices

```
all_preds = [
    ('VADER\n(Lexicon)',            y_pred_vader,   'Oranges'),
    ('TF-IDF +\nLogistic Reg',      y_pred_lr,      'Blues'),
    ('Hybrid SVM\n+ Lexicon',       y_pred_svm,     'Purples'),
    ('FinBERT\nBaseline',           y_pred_fb_base, 'YlOrRd'),
    ('FinBERT\nFinetuned ⋆',        y_pred_fb_ft,   'Greens'),
]

fig, axes = plt.subplots(1, 5, figsize=(22, 4))
fig.suptitle('Confusion Matrices — All Models on Same Test Set',
fontsize=14, fontweight='bold', y=1.02)

for ax, (name, y_pred, cmap) in zip(axes, all_preds):
    cm = confusion_matrix(y_test, y_pred, labels=LABELS)
    acc = accuracy_score(y_test, y_pred)

    sns.heatmap(
        cm, annot=True, fmt='d', cmap=cmap,
        xticklabels=['neg', 'neu', 'pos'],
        yticklabels=['neg', 'neu', 'pos'],
        linewidths=0.5, linecolor='grey',
        ax=ax, cbar=False
    )
    ax.set_title(f'{name}\nAcc: {acc*100:.1f}%', fontsize=10,
fontweight='bold')
    ax.set_xlabel('Predicted', fontsize=9)
    ax.set_ylabel('True', fontsize=9)

plt.tight_layout()
plt.show()
print('Saved as confusion_matrices_all.png')
```




```
Saved as confusion_matrices_all.png
```

```
import zipfile
import os

# Unzip first
with zipfile.ZipFile('finbert_finetuned.zip', 'r') as zip_ref:
    zip_ref.extractall('.')
```

```python
print('□ Unzipped!')

# Then load normally
import torch
from transformers import AutoTokenizer,
AutoModelForSequenceClassification, pipeline

device = 0 if torch.cuda.is_available() else -1

tokenizer = AutoTokenizer.from_pretrained('finbert_finetuned')
model =
AutoModelForSequenceClassification.from_pretrained('finbert_finetuned'
)

label_map = {0: 'neutral', 1: 'negative', 2: 'positive'}
model.config.id2label = label_map
model.config.label2id = {v: k for k, v in label_map.items()}

classifier = pipeline('text-classification', model=model,
tokenizer=tokenizer, device=device)

print('□ Model loaded! Type your sentence below.')

# ── Type your own sentence ──────────────────────────────────────
while True:
    text = input('\nEnter a sentence (or type quit to stop): ')
    if text.lower() == 'quit':
        break
    result = classifier(text, truncation=True, max_length=512)[0]
    print(f'Sentiment  : {result["label"].upper()}')
    print(f'Confidence : {result["score"]*100:.1f}%')
```

```
□ Unzipped!
```

{"model_id":"939d85a4f8e4442d8f7bce6b88b20e05","version_major":2,"version_minor":0}

```
□ Model loaded! Type your sentence below.

Enter a sentence (or type quit to stop): War left Iran in ruin
Sentiment  : NEUTRAL
Confidence : 99.0%

Enter a sentence (or type quit to stop): quit
```